

Module 3

Chapter 1

Introduction to Multivariable Calculus

We now enter a brief module where we have one mathematical goal in mind, which is being able to find a ***gradient*** of a multivariable function, and one application in mind, which is to understand the machinery of one of the most popular modern machine learning methods: ***Gradient Descent***. At the end of this module we will also have a chapter dedicated to the review of Integrals which will be important to understand concepts of Statistics which we will be learning about in our next module.

Hill Climbing

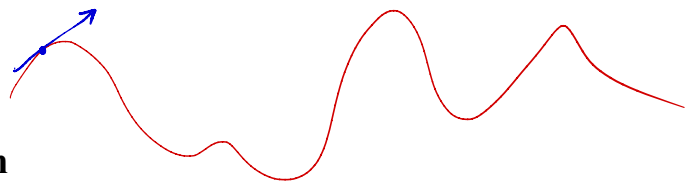
Consider the scenario: You are on a cross country tour and take a pitstop at a dodgy hotel for the night. The internet reception at the hotel is poor and all you have for you entertainment, is the television set in the room. But there's a problem: The remote control for the TV has worn out buttons and you cannot understand all the buttons on it. You start hitting random buttons on the remote control and observing whether the situation becomes better or worse as a result. You'll sometimes make the situation worse, like making the volume louder than you intended or changing the channel when you didn't mean to, but once you find the right button for doing something like turning up the volume, you'll hit it a lot and get to the state you want.

"Hill climbing" is the general term for such an approach; we can visualize the situation as being like a 3D graph, where the ***inputs*** make up 2 axes and the ***goodness*** of the situation is plotted on the third axis. We may not know the whole

landscape of goodness, but we can try “hitting buttons” (*adjusting parameters*) and seeing whether the situation improves as a result. If we keep at this, we will eventually reach a *local optimum* in the goodness of the situation. However, we *aren’t guaranteed of a global optimum*, because we only sample states that are very similar to the current one. Hill climbing will not find a great program 10 channel-changes away; it checks what happens for small perturbations of variables, and doesn’t go down routes that don’t look immediately better.

Hill climbing can be appropriate when we have no insight into the underlying function, or it’s not differentiable. But if we know how the goodness of the situation varies as a *continuous function* of the underlying *parameters*, we can do better than hill-climbing, by finding the *gradient* of the function. The gradient is simply a *vector* in the direction of the *slope* of a surface - the *multivariable* version of the *derivative*. This vector points in the *most uphill* direction possible, a direction we may not have thought to sample if we were hill climbing. So if the *function* represents *goodness*, we want to follow that vector straight *uphill* when we adjust all our parameters to maximize it. And if it represents *error*, we want to flip the direction of the vector and head in the exact opposite direction, i.e. *downhill* to minimize it. This is the basic idea behind “*gradient descent*.” Like hill climbing, it has the potential to become trapped in local minima (if heading downhill) or maxima (if heading uphill). But for a variety of very complex functions to optimize, this somehow doesn’t seem like a huge problem in practice, for a wide variety of problems.

Understanding gradient descent requires the knowledge of Gradient, and understanding the Gradient, requires knowledge of Calculus. In this chapter we will have a quick review of the basic concepts of Multivariable Calculus (esp. derivatives of a function).



Understanding the derivative of a function

Given a single variable function, $f(x)$, the derivative of the function can be written as $f'(x)$ or $\frac{df}{dx}$. The derivative of a function at any point represents the **slope of the tangent** to the function at that point. It can also be seen as an instantaneous **rate of change** for each point, with the connection that the **tangent** line provides a **linear approximation** to the function at each point. Thus, for example, if we have a function $x(t)$ for position at each time t , then $x'(t)$ represents the instantaneous **velocity** at each time, and $x''(t)$ represents the instantaneous **acceleration**, or rate of change in the rate of change.

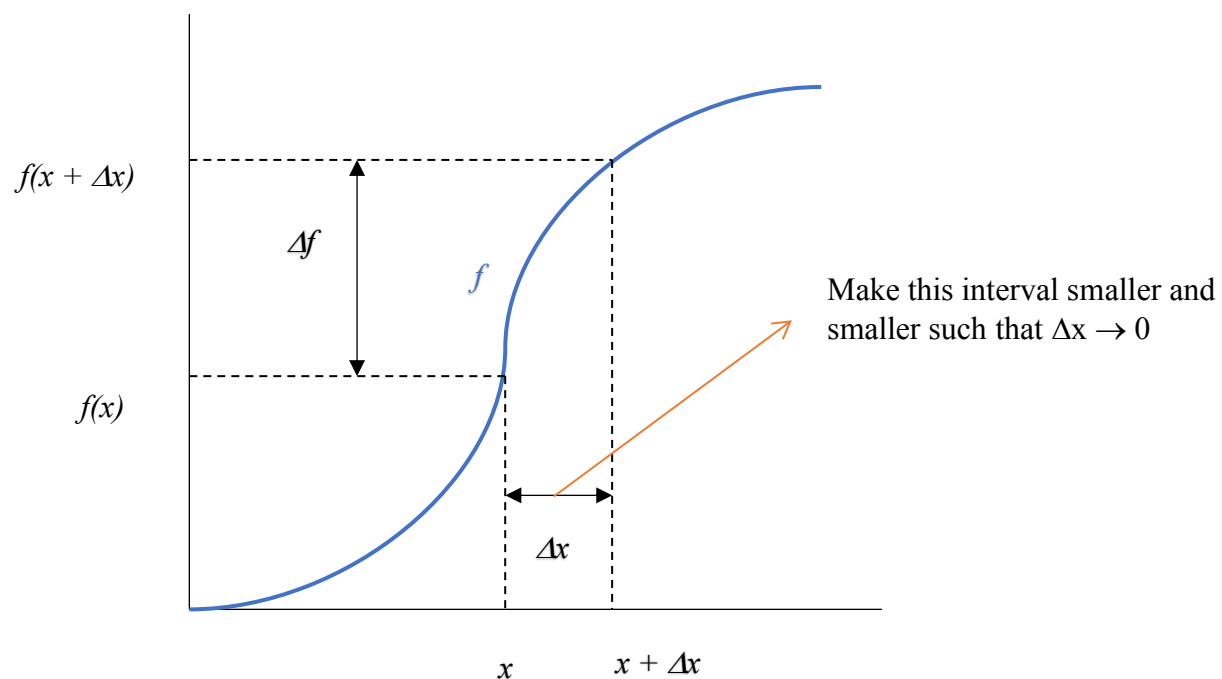


Fig 1

$$\text{Derivative of } f(x) = f'(x) = \frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

In Fig 1, Δx represents a **change** in the value of x . We keep making the interval between x and $(x+\Delta x)$ smaller and smaller until it is infinitesimal (**limit** $\Delta x \rightarrow 0$). The numerator $f(x+\Delta x)-f(x)$ represents the corresponding **change** in the value of the function f over the **interval** Δx . This makes the derivative of a function f at a point x , the **rate of change of f** at that point.

However, a function is not always differentiable. Following are the few cases where the derivative of a function at a point **does not** exist:

1. If the function is not defined at a point
2. Function does not have a limit at that point
3. If the function is not continuous at a point
4. The function has a sudden jump at a point

Basic rules for finding the derivatives of a single variable function

- **Trigonometry rules:**

1. When $f(x) = \sin(x)$, $f'(x) = \cos(x)$
2. When $f(x) = \cos(x)$, $f'(x) = -\sin(x)$
3. When $f(x) = \tan(x)$, $f'(x) = \sec^2(x)$

- **Power Rule:** When $f(x) = x^n$, $f'(x) = nx^{n-1}$

For example: $f(x) = x^4$, $f'(x) = 4x^3$

- **Multiplication by a constant:** When $f(x) = cg(x)$, $f'(x) = cg'(x)$

For example: $f(x) = 3x^4$, $f'(x) = 3*4x^3 = 12x^3$

- **Sum/difference Rule:** When $f(x) = g(x) \pm h(x)$, $f'(x) = g'(x) \pm h'(x)$

For example: $f(x) = x^2 + x^3$, $f'(x) = 2x + 3x^2$

- **Product Rule:** When $f(x) = g(x)h(x)$, $f'(x) = g(x) h'(x) + h(x) g'(x)$

For example: $f(x) = \sin(x)\cos(x)$, $f'(x) = \sin(x)(-\sin(x)) + \cos(x)\cos(x) = \cos^2(x) - \sin^2(x)$

- **Quotient Rule:** When $f(x) = \frac{g(x)}{h(x)}$, $f'(x) = \frac{g'(x)h(x) - h'(x)g(x)}{h^2(x)}$

For example: $f(x) = \frac{\cos(x)}{x}$, $f'(x) = \frac{-\sin(x) \cdot x - 1 \cdot \cos(x)}{x^2} = -\frac{\cos(x) + x\sin(x)}{x^2}$

- **Reciprocal Rule:** When $f(x) = \frac{1}{g(x)}$, $f'(x) = \frac{-g'(x)}{g^2(x)}$

For example: $f(x) = \frac{1}{x}$, $f'(x) = \frac{-1}{x^2}$

- **Chain Rule:** When $f(x) = h(g(x))$, $f'(x) = h'(g(x))g'(x)$

For example: $f(x) = \sin(x^2)$, $f'(x) = \cos(x^2) \cdot 2x = 2x \cos(x^2)$

Ext $\rightarrow \sin(x)$ $\cos(x^2) \cdot 2x$
 Int $\rightarrow \textcircled{x^2}$ $= 2x \cos(x^2)$

Let's Practice

Find the derivatives of the following:

- $f(x) = 6x^3 - 9x + 4$
- $y = 2t^4 - 10t^2 + 13t$
- $f(x) = \frac{x^2 + 2x + 1}{x}$
- $f(x) = (2x + 1)^3$
- $f(x) = \frac{5}{(1-x)}$
- $f(x) = 3x^2(1 + x)^2$

Find the slope of the tangent to the given function at $x = -1$:

$y = 7x^4 + 8x^{-6} + 2x$

$$1. \quad 6x^3 - 9x + 4$$

$$\rightarrow 18x^2 - 9$$

$$2. \quad 2t^4 - 10t^2 + 13t$$

$$\rightarrow 8t^3 - 20t + 13$$

$$3. \quad \frac{x^2 + 2x + 1}{x}$$

OR $\rightarrow f(x) = (x + 2 + x^{-1}) \quad f'(x) = 1 - \frac{1}{x^2}$

$$\rightarrow \left\{ f(x) = \frac{g(x)}{h(x)} \quad \therefore f'(x) = \frac{g'(x) \cdot h(x) - h'(x) \cdot g(x)}{h^2(x)} \right\}$$

$$\frac{(2x + 2)x - (x^2 + 2x + 1)}{x^2}$$

$$= \frac{2x^2 + 2x - x^2 - 2x - 1}{x^2}$$

$$= \frac{x^2 - 1}{x^2} = 1 - \frac{1}{x^2}$$

4. $(2x+1)^3$

$$\left\{ \begin{array}{l} f(x) = h(g(x)) \quad \therefore f'(x) = h'(g(x)) \cdot g'(x) \\ \text{here, } h(x) = x^3 \quad g(x) = (2x+1) \end{array} \right\}$$

$$\therefore h'(g(x)) = 3(2x+1)^2$$

$$g'(x) = 2$$

$$\therefore f'(x) = 6(2x+1)^2$$

5. $\frac{5}{(1-x)}$

$$\left\{ \begin{array}{l} f(x) = \frac{g(x)}{h(x)} \quad \therefore f'(x) = \frac{g'(x) \cdot h(x) - h'(x) \cdot g(x)}{h^2(x)} \end{array} \right\}$$

$$f'(x) = \frac{0 - (-1) \cdot 5}{(1-x)^2} = \frac{5}{(1-x)^2}$$

6. $3x^2(1+x)^2$

$$\left\{ \begin{array}{l} f(x) = g(x)h(x) \quad \therefore f'(x) = g(x)h'(x) + h(x)g'(x) \end{array} \right\}$$

here

$$g(x) = 3x^2 \quad \therefore g'(x) = 6x$$

$$h(x) = (1+x)^2$$

$$h'(x) = 2(1+x)$$

using chain rule:

$$\begin{aligned}
 \therefore f'(x) &= 6x^2(1+x) + 6x(1+x)^2 \\
 &= (1+x)(6x^2 + 6x(1+x)) \\
 &= (1+x)(6x^2 + 6x + 6x^2) \\
 &= (1+x)(12x^2 + 6x) \\
 &= 6x(1+x)(2x+1)
 \end{aligned}
 \quad \left. \vphantom{\begin{aligned} f'(x) &= 6x^2(1+x) + 6x(1+x)^2 \\ &= (1+x)(6x^2 + 6x(1+x)) \\ &= (1+x)(6x^2 + 6x + 6x^2) \\ &= (1+x)(12x^2 + 6x) \\ &= 6x(1+x)(2x+1) \end{aligned}} \right\} \text{optional}$$

→ Given function: $y = 7x^4 + 8x^{-6} + 2x$

slope of a tangent is given by the derivative at that point

$$\therefore \frac{dy}{dx} = 28x^3 - 48x^{-7} + 2$$

$$\left. \frac{dy}{dx} \right|_{x=-1} = 28(-1)^3 - 48(-1)^{-7} + 2$$

$$= -28 + 48 + 2$$

$$= 50 - 28 = 22$$

Derivatives with respect to particular variables for multivariable functions

We now move on to the case where we have a function with multiple variables.

Suppose we have a ***multivariable*** function $f(x,y)$, and we want to take a derivative. Since the inputs are two-dimensional, the question arises as to which direction we want a measure of the slope in, since it could be different in different directions. In fact, if we're going to allow more than one of the variables to change, there are then going to be an infinite amount of ways for them to change. For instance, one variable could be changing faster than the other variable(s) in the function. It is also possible for the function to be changing differently depending on how we allow one or more of the variables to change.

In this chapter we are going to concentrate exclusively on only changing one of the variables at a time, while the remaining variable(s) are held fixed.

The key here, is to remember that one variable, ***the one we are taking the derivative with respect to***, is allowed to ***vary***, but the others are not (they are treated like any other constant). This is a rough explanation of how ***partial derivatives*** work.

Since partial derivatives depend on the variable we are finding the derivative with respect to, the ***notation*** needs to be specific. For example, if we are finding the ***partial derivative*** of a function $f(x,y)$ with respect to x , then the notation is:

$$f_x(x,y) \text{ or } \frac{\partial f(x,y)}{\partial x}.$$

Let's take an example

$$\text{Let } f(x,y) = x^2 + y^2 \quad \begin{array}{l} f_x = 2x + 0 \\ \quad = 2x \\ f_y = 0 + 2y \\ \quad = 2y \end{array}$$

Then the partial derivative w.r.t x is

$$f_x(x, y) = 2x$$

And the partial derivative w.r.t y is

$$f_y(x, y) = 2y$$

******Notice the “partial derivative” notation – partial since it is only with respect to one of multiple variables. The curly ∂ is a difference from the notation for single-variable calculus. When we can take partial derivatives with respect to multiple variables, the result gives us the relative strength of effect that each variable has on the output, in a particular region of the input space. A large derivative (positive or negative) means the output is very sensitive to the variable, and zero would mean that adjusting that variable has no effect.

Let's practice

Find f_x and f_y

$$f(x, y) = \frac{x^2}{y^2 + 1}$$

Find f_y

$$f(x, y) = x^3 - 2x^2 + x$$

Find f_x

$$f(x, y) = \frac{y^2}{2} + x^3$$

Find f_x and f_y

- $f(x, y) = (x^2 - 1)(y + 2)$
- $(3x - y)^5$

$$1. f(x, y) = \frac{x^2}{y^2 + 1}$$

$$f_x(x, y) = \frac{2x}{y^2 + 1}$$

using quotient rule

$$\begin{aligned} f_y(x, y) &= \frac{0 - 2yx^2}{(y^2 + 1)^2} \\ &= -\frac{2x^2y}{(y^2 + 1)^2} \end{aligned}$$

$$2. f(x, y) = x^3 - 2x^2 + x$$

$$f_y = 0$$

$$3. f(x, y) = \frac{y^2}{2} + x^3$$

$$\rightarrow f_x = 3x^2$$

$$4) 1) f(x, y) = (x^2 - 1)(y + 2)$$

$$f_x = (y + 2)(2x) = 2xy + 4x$$

$$f_y = (x^2 - 1)(1) = (x^2 - 1)$$

$$4) 2) (3x - y)^5$$

using chain rule

$$f_x = 5(3x - y)^4 \cdot 3 = 15(3x - y)^4$$

$$f_y = 5(3x - y)^4 \cdot (-1) = -5(3x - y)^4$$

Second Order Derivative

Second order derivatives are essentially *derivatives of derivatives*. In the *single-variable* case, there's just *one* possible second order derivative. But with *multiple variables*, when we take the derivative again, it could be with respect to the *same variable*, or a *different* one. We can indicate these second derivatives with either *subscripts* that list the variables in order of differentiation or using the *partial derivative notation* where the *earlier* derivatives are to the *right* of the later ones. In the latter notation, we can group identical derivative operators with squares:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} \text{ (2nd derivative in x direction)}$$

$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} \text{ (derivative w.r.t x, then y)}$$

Let's Practice

Find all the second order partial derivatives of:

- $f(x) = 3x^4 - 2x^3 + 4x^2 - 5x + 1$
- $f(x) = \sqrt[4]{x+1}$
- $f(x, y) = x^2y - y^3x^4$

Understanding a few key terms: Maxima, Minima, Saddle Points

One of the most important applications of calculus is its ability to sniff out the maximum or the minimum of a function. Let's understand how!

$$1. \quad 3x^4 - 2x^3 + 4x^2 - 5x + 1$$

$$f_x \text{ or } f'(x) = 12x^3 - 6x^2 + 8x - 5$$

$$f_{xx} \text{ or } f''(x) = 36x^2 - 12x + 8$$

$$2. \quad f(x) = \sqrt[4]{(x+1)} = (x+1)^{1/4}$$

Using chain rule:

$$f_x = \frac{1}{4} (x+1)^{1/4-1} = \frac{1}{4} (x+1)^{-3/4}$$

$$\begin{aligned} f_{xx} &= \frac{1}{4} \times \left(-\frac{3}{4}\right) (x+1)^{-3/4-1} \\ &= -\frac{3}{16} (x+1)^{-7/4} = -\frac{3}{16 \sqrt[4]{(x+1)^7}} \end{aligned}$$

$$3. \quad f(x, y) = x^2y - y^3x^4$$

$$f_x = 2xy - 4x^3y^3$$

$$f_y = x^2 - 3y^2x^4$$

$$f_{xx} = 2y - 12x^2y^3$$

$$f_{yy} = -6yx^4$$

$$f_{xy} = 2x - 12x^3y^2$$

$$f_{yx} = 2x - 12x^3y^2$$

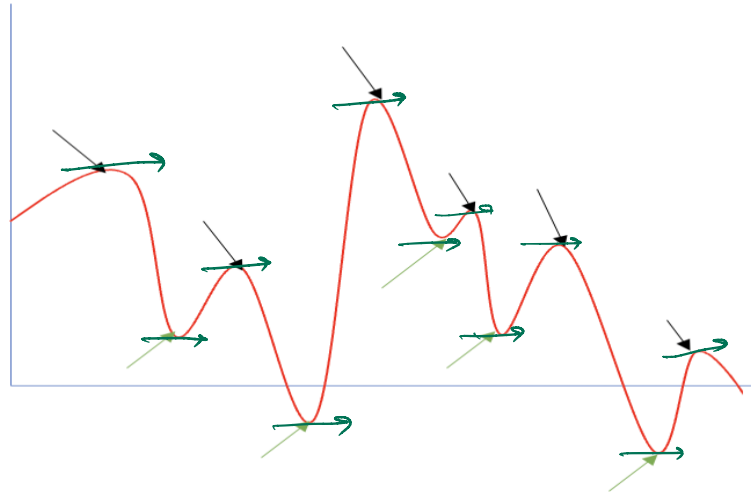


Fig 2

Let us look at the above fig 2. Here we have the graph of a function with a lot of bumps and peaks. All the peaks in the function are denoted by the **black arrows**. Each of these peaks is called a **Local Maximum** and the plural is called **Local Maxima**. The **input**, say, (x_0, y_0) , at these peaks are called the **Local Maximum points** while the **output/value** of the function at these points is the **Local Maximum** itself.

** The word "local" is used to distinguish these from the **global maximum** of the function, which is the single greatest value that the function can achieve. If you are on the peak of a mountain, it's a local maximum, but unless that mountain is Mt. Everest, it is not a global peak. Intuitively, it is a special point in the input space where taking a small step in any direction can only decrease the value of the function.

Similarly, the **inverted peaks** in the graph are denoted by the **green arrows** and are called **Local Minima**, with the singular being **Local Minimum**. The **input**, say, (x_0, y_0) , at these inverted peaks are called the **Local Minimum points** while the **output/value** of the function at these points is the **Local Minimum** itself.

In general, **local maxima and minima** of a function f are studied by looking for input values a where $f'(a) = 0$ (Fig 3 a). This is because as long as the function is continuous and differentiable, the tangent line at peaks and valleys will flatten out, in that it will have a slope of 0.

Such a point a has various names:

- **Stable point**
- **Critical point**
- **Stationary point**

For this derivative test to work, a function needs to be continuous and differentiable. However, even when f is continuous and differentiable, it is not enough for the derivative to be 0 to be a local maximum or minimum, it can also happen at **inflection or saddle points** (Fig 3 b).

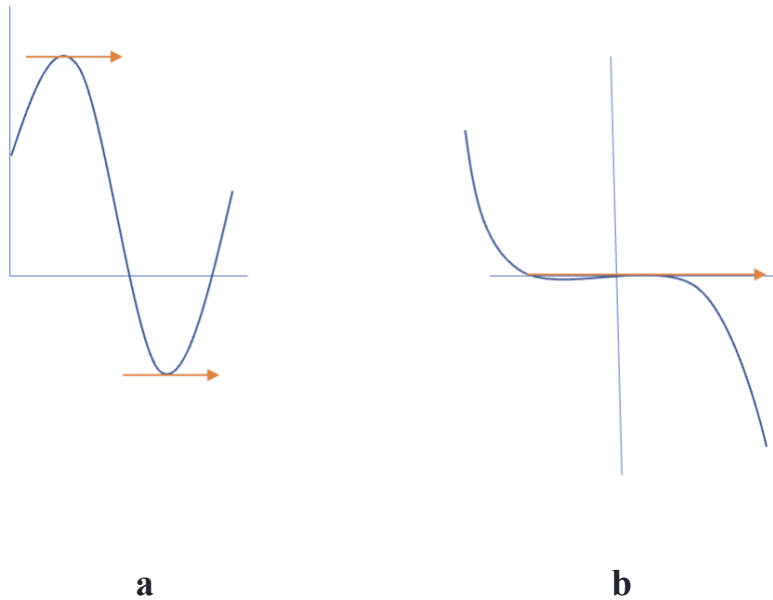


Fig 3

This means finding stable points is a good way to start the search for a maximum, but it is not necessarily the end.

The story is very similar for multivariable functions. When the function is continuous and differentiable, ***all the partial derivatives will be 0 at a local maximum or minimum point.*** As with single variable functions, **it is not enough** to ensure that a point is a local maximum or minimum.

How to find Local Maximum/Minimum points and Saddle points: Second order Derivative test

Suppose you have a function $f(x,y)$ and you found the critical point at (x_0, y_0) (using the first order derivative rule). We define D as:

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2$$

- If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then (x_0, y_0) is a Local Maximum point and $f(x_0, y_0)$ is ***Local Maximum***
- If $D > 0$ and $f_{xx}(x_0, y_0) > 0$ then (x_0, y_0) is a Local Minimum point and $f(x_0, y_0)$ is ***Local Minimum***
- If $D < 0$ then (x_0, y_0) is a ***Saddle Point***
- If $D = 0$, then the test is ***inconclusive***

Let's take an example

Let's find the local maximum, minimum and saddle points for the function:

$$f(x, y) = 12x^2 + y^3 - 12xy$$

Step 1:

Find the ***critical points***:

Points at which all partial derivatives are 0:

$$f_x = 24x - 12y = 0$$

$$f_y = 3y^2 - 12x = 0$$

This gives us:

$$y = 2x$$

$$y^2 = 4x^2$$

That gives us $x = 0, 1$

When $x = 0, y = 0$

When $x = 1, y = 2$

So, our Critical Points are $(0,0)$ and $(1,2)$

Step 2:

Calculate D:

$$f_{xx} = 24$$

$$f_{xy} = -12$$

$$f_{yy} = 6y$$

$$D = 144y - 144$$

$$\text{Thus, } D(0,0) = 144 \cdot 0 - 144 = -144$$

$$\text{And } D(1,2) = 144 \cdot 2 - 144 = 144$$

Since $D(0,0) < 0$, $(0,0)$ is a **Saddle Point**

Since $D(1,2) > 0$ and $f_{xx}(1,2) > 0$, $(1,2)$ is a **local Minimum point**

Let's Practice

Find the critical points of the given function and check if they are saddle points, local maximum points or minimum points:

- $f(x, y) = y^2 - x^2$
- $f(x, y) = x^2y^2 - 5x^2 - 5y^2 - 8xy$

Q. Find the critical points of the given function $f(x,y) = y^2 - x^2$ and check for the presence of any saddle points.

Finding critical points:

$$\rightarrow f_x(x,y) = -2x$$

$$f_y(x,y) = 2y$$

$$\left. \begin{array}{l} f_x(x,y) = 0 \text{ gives } x=0 \\ f_y(x,y) = 0 \text{ gives } y=0 \end{array} \right\} \rightarrow \begin{array}{l} -2x=0 \\ 2y=0 \\ x=0 \\ y=0 \end{array}$$

$\therefore$ Critical point $\Rightarrow (0,0)$

Checking for saddle points:

$$f_{xx}(x,y) = \frac{\partial(-2x)}{\partial x} = -2$$

$$f_{yy}(x,y) = \frac{\partial(2y)}{\partial y} = 2$$

$$f_{xy}(x,y) = \frac{\partial(-2x)}{\partial y} = 0$$

$$\begin{aligned} \therefore D &= f_{xx}(0,0)f_{yy}(0,0) - [f_{xy}(0,0)]^2 \\ &= -4 - 0^2 = -4 \end{aligned}$$

$$\Rightarrow D < 0$$

$\therefore$ at $(0,0)$ f has a saddle point

Q. Find the critical points of the function

$$f(x, y) = x^2 y^2 - 5x^2 - 5y^2 - 8xy$$

and check for the presence of any saddle points

$$\rightarrow f_x(x, y) = 2xy^2 - 10x - 8y$$

$$f_y(x, y) = 2x^2 y - 10y - 8x$$

$$2xy^2 - 10x - 8y = 0 \quad \text{---(i)}$$

$$2x^2 y - 10y - 8x = 0 \quad \text{---(ii)}$$

subtracting (ii) from (i),

$$2xy^2 - 2x^2 y - 2x + 2y = 0$$

$$2xy^2 - 2x^2 y - 10x + 10y - 8y + 8x = 0 \quad y = -1/x$$

$$2xy(y-x) + 2(y-x) = 0 \rightarrow (2xy+2) = 0 \text{ or } (y-x) = 0 \rightarrow y = x$$

$$\Rightarrow y = x \text{ or } y = -1/x$$

If $y = x$, substituting either in f_x or f_y , we get

$$2x^3 - 18x = 0 \Rightarrow x = 0, 3, -3$$

$\therefore$ our points are: $(0, 0), (3, 3), (-3, -3)$

If $y = -1/x$, substituting either in f_x or f_y , we get

$$\frac{2}{x} - 10x + \frac{8}{x} = 0 \Rightarrow 1 - x^2 = 0 \Rightarrow x = 1, -1$$

$\therefore$ our points are $(1, -1), (-1, 1)$
 $= (1, -1) \quad (-1, 1)$

∴ Our critical points are: $(0,0)$, $(3,3)$, $(-3,-3)$, $(1,-1)$ and $(-1,1)$

→ Calculating D :

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$$
$$f_{xx}(x, y) = 2y^2 - 10$$
$$f_{yy}(x, y) = 2x^2 - 10$$
$$f_{xy}(x, y) = 4xy - 8$$
$$\therefore D = (2y^2 - 10)(2x^2 - 10) - (4xy - 8)^2$$

At $(0,0)$, $D = 36$, $f_{xx}(0,0) = -10$
 $\Rightarrow D > 0$, $f_{xx} < 0$
∴ at $(0,0)$ we have local maximum

At $(\pm 3, \pm 3)$, $D = -720 < 0 \Rightarrow$ at $(\pm 3, \pm 3)$ we have saddle points
At $(\pm 1, \pm 1)$, $D = -80 < 0 \Rightarrow$ at $(\pm 1, \pm 1)$ we have saddle points

∴ $(0,0) \rightarrow$ local Max

$(3,3), (-3,-3), (1,-1), (-1,1) \rightarrow$ saddle points